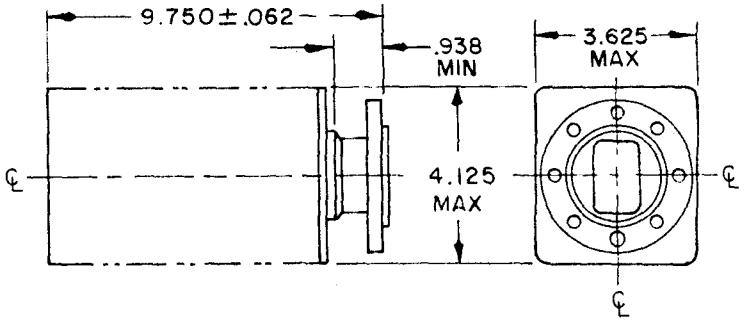
MIL-D-3954/17A
29 May 1974
SUPERSEDING
MIL-D-3954/17
14 October 1966, and
MS35892
15 May 1962

MILITARY SPECIFICATION SHEET

DUMMY LOADS, ELECTRICAL, WAVEGUIDE (FREQUENCY RANGE 3.95 TO 5.85 GIGAHERTZ)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dummy loads described herein shall consist of this document and the latest issue of Specification MIL-D-3954.



INCHES	MM
.062	1.57
.938	23.83
3.625	92.08
4.125	104.78
9.750	247.65

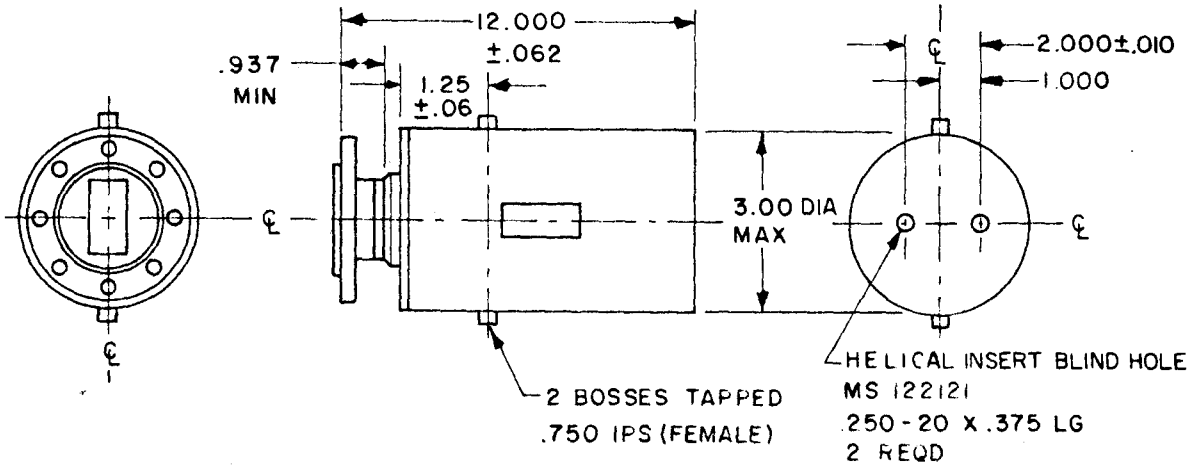
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.
5. Quantity of fins required will be dependent on heat level requirement of the load.

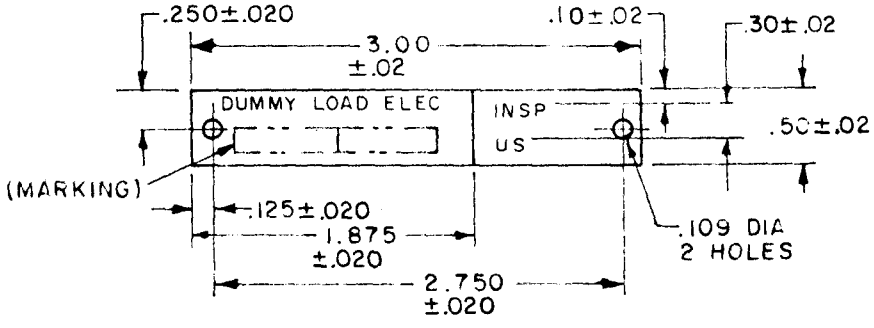
Ⓐ FIGURE 1. Class I dummy load.

Ⓐ denotes changes.

FSC 5985

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IDENTIFICATION PLATE

INCHES	MM	INCHES	MM	INCHES	MM
.010	.25	.250	6.35	1.25	31.75
.02	.51	.30	7.62	1.875	47.63
.06	1.52	.375	9.53	2.000	50.80
.062	1.57	.50	12.70	2.750	69.85
.10	2.54	.750	19.05	3.00	76.20
.109	2.77	.937	23.80	12.000	304.80
.125	3.18	1.000	25.40		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.

Ⓐ FIGURE 2. Class II dummy load.

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TABLE I. Characteristics.

Part number	Class Ⓐ	Material	Flange equal to Ⓐ	VSWR max	Power		Pressure		Figure	Flow rate	Input temp.	Output temp.
					Average	Peak	Internal	Coolant chamber				
					(watts)	(kilowatts)	(psig)	(psig)		(GPM 1/)	(°C)	(°C)
M3954/17-01	I	AL	M3922/57-001 (UG-407/U)	1.10	1000 2/	1000	17.2	-	1	-	-	-
M3954/17-02	II	AL	M3922/57-001 (UG-407/U)	1.10	4000	1300	30	100	2	1.25	35	60
M3954/17-03	II	Copper	M3922/57-002 (UG-149/U)	1.10	4000	1300	30	100	2	1.25	35	60
M3954/17-04	II	Corrosion resisting steel	M3922/57-002 (UG-149/U)	1.10	4000	1300	39	100	2	1.25	35	60

1/ The flow rate was determined from the following formula:

$$Q = \frac{6.8P}{Cp\ \Delta T}$$

Where: Q = Minimum flow rate in GPM.
P = Avg power in kilowatts.
Cp = Specific heat of coolant.
ΔT = Coolant temperature rise in °F.

The calculations were made for Cp = 1 for water, a ΔT of 45°F and a safety factor of approximately 2. For different coolants or different temperature rises, a different flow rate would be necessary.

2/ This is a test power value. The average rated power is 2,000 watts.

Ⓐ TABLE II. Cross reference part number/ AN nomenclature.

Part number	AN nomenclature
M3954/17-01	DA-149/U

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REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figures 1 and 2.
Weight:
Dry load - 6.4 lbs max.
Liquid cooled load - 13 lbs max.

Performance characteristics: See table I.

Part number: M3954/17- (dash number from table I).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - CG, MC, AS, OS, SH
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0754-7)

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